

Ryan Cho

Senior Backend Software Engineer

+1 608 360 1459 · Ryanmcho100@gmail.com · <https://www.linkedin.com/in/ryan-cho-2020/> · <https://github.com/Chomama> · New York, United States

PROFESSIONAL SUMMARY

Software engineer with 7+ years building scalable backend systems, APIs, and cloud infrastructure across fintech and enterprise platforms. Deep expertise in Python, service-oriented architecture, and security-conscious design for high-availability products serving large user bases. Proven track record delivering performant integrations, strengthening internal tooling, and collaborating across engineering, product, and growth teams to ship intuitive, production-ready solutions.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Architected and maintained **Python backend services** and client-facing APIs for portfolio analytics, supporting high-volume daily workflows across distributed microservices with consistent sub-200ms p95 latency.
- Led design and delivery of **RESTful API integrations** with third-party data providers and internal research pipelines, enabling reliable ingestion of market signals into real-time decision engines.
- Strengthened **AWS cloud infrastructure** with Docker and Kubernetes orchestration, improving deployment reliability and reducing incident recovery time through automated rollback and health-check policies.
- Built **payment and billing workflows** using Stripe webhooks and reconciliation jobs, ensuring PCI-aligned handling of subscription lifecycle events across client accounts.
- Implemented **comprehensive automated testing** with pytest, contract tests, and CI quality gates, raising backend coverage and catching regressions before production releases.
- Drove **security hardening initiatives** including secrets rotation, OAuth2 validation, and OWASP-aligned input sanitization across all external-facing endpoints.
- Partnered with frontend and product teams to ship **AI-assisted analytics features**, translating model outputs into stable API contracts consumed by React dashboards.
- Optimized **PostgreSQL and Redis caching layers** for high-read portfolio workloads, reducing database load while preserving consistency across concurrent updates.
- Mentored junior engineers on **service design patterns**, code review standards, and observability practices, accelerating onboarding and improving release readiness.

Technology Analyst, Incoming Technology Analyst, Technology

May/2019 - Apr/2022

Summer Analyst,

BNP Paribas

Remote | Seventh Avenue, New York

- Developed **Python and Java backend components** for trade reporting and regulatory workflows, processing large daily record volumes across globally distributed banking systems.
- Designed **REST and message-queue APIs** connecting front-office trading platforms with middle-office validation services, improving data lineage traceability for audit requirements.
- Contributed to **AWS cloud migration initiatives**, containerizing legacy batch jobs and establishing infrastructure-as-code templates for repeatable environment provisioning.
- Enforced **security fundamentals** including role-based access controls, encryption at rest, and secure credential management aligned with financial industry compliance standards.
- Built **automated test frameworks** for API contract validation and regression suites, reducing production defects across consecutive release cycles.
- Collaborated with cross-functional stakeholders to **translate business requirements** into scalable system designs, balancing performance constraints with regulatory deadlines.
- Optimized **data structures and storage strategies** for time-series market data, enabling faster query performance on multi-terabyte historical trade archives.

- Supported **on-call rotation** for critical payment and settlement pipelines, diagnosing latency spikes and implementing targeted fixes under strict SLA windows.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Built **full-stack features** for a restaurant technology platform using React and Node.js, improving order tracking visibility for venue partners across the network.
- Implemented **REST API endpoints** for menu synchronization and inventory updates, applying consistent error-handling and pagination patterns across services.
- Wrote **unit and integration tests** with automated CI coverage, contributing to measurable gains in backend test coverage during the internship tenure.
- Participated in **Agile sprint ceremonies** and code reviews, delivering incremental backend enhancements aligned with product roadmap priorities.
- Documented **API schemas and deployment runbooks**, enabling smoother handoffs between intern cohorts and full-time engineering staff.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Developed **responsive web applications** for nonprofit data visualization initiatives using JavaScript, HTML, and CSS, supporting community outreach programs across New York.
- Integrated **third-party APIs** for donation processing and newsletter subscriptions, establishing reusable fetch utilities and client-side validation patterns.
- Collaborated with foundation stakeholders to **translate design mockups** into accessible, mobile-friendly pages deployed on cloud-hosted static infrastructure.
- Established **basic CI workflows** for linting and build verification, reducing manual QA steps before each release cycle.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: React, TypeScript, JavaScript, HTML5, CSS3, Responsive Web Design

Backend: Python, FastAPI, Django, REST APIs, Microservices, Node.js, System Design Patterns

Databases: PostgreSQL, Redis, MongoDB, SQL, Data Modeling, Query Optimization

Cloud & DevOps: AWS, Kubernetes, Docker, Terraform, CI/CD, Infrastructure as Code

Monitoring & Tools: Datadog, Prometheus, Git, pytest, Postman, Stripe API

Workflow & PM: Agile/Scrum, Jira, Cross-functional Collaboration, Technical Documentation, Code Review